Oscillations of a Viscous Liquid Drop

A Self-Contained Pedagogical Expansion of Reid (1960)

Expanded from W. H. Reid, *Quart. Appl. Math.* **18**(1), 86–89 (1960)

Contents

1 Introduction and Physical Motivation

1.1 What problem are we solving?

Consider a small liquid droplet (say, a water drop suspended in air) that has been slightly deformed from its equilibrium spherical shape. Surface tension acts as a restoring force, and the drop will oscillate. Viscosity damps those oscillations. The question is: *at what frequency does it oscillate, and how fast does it decay?*

This couples three distinct pieces of physics:

1. The inviscid capillary (surface-tension-driven) oscillation modes, characterized by the frequency $\sigma_{l;0}$ for a deformation of spherical-harmonic order l.
2. Viscous dissipation inside the bulk of the drop.
3. The coupling between surface deformation, internal flow, and the pressure field, all of which must be consistent at the boundary.

Reid's paper resolves all three simultaneously for *arbitrary* viscosity, by reducing the problem to a single transcendental characteristic equation. Its central insight is a rescaling. Surface tension enters only through the inviscid frequency $\sigma_{l;0}$. After non-dimensionalisation, the characteristic equation is identical to Chandrasekhar's equation for a self-gravitating viscous liquid globe.

1.2 Historical context and relation to Molaček & Bush (2012)

Three analytical milestones precede Reid's result:

- **Rayleigh (1879) / Lamb (1932)**: inviscid frequencies, Eq. (13).
- **Lamb (1881)**: small-viscosity correction, $\sigma_{l;\nu} = (l-1)(2l+1)\nu/R^2 \pm \mathrm{i}\,\sigma_{l;0}$.
- **Chandrasekhar (1959)**: full arbitrary-viscosity solution for a *self-gravitating* globe.

Reid (1960) shows the surface-tension problem is identical to Chandrasekhar's, thereby completing the picture for surface-tension-driven drops.

The result is directly used in Molaček & Bush (2012), where the coefficients A_m and D_m governing kinetic energy and viscous dissipation for each surface mode are extracted from Reid's characteristic equation (Eq. (39) below).

2 Mathematical Preliminaries

2.1 Spherical harmonics

A *spherical harmonic* of degree l and order m is the function

$$Y_l^m(\theta, \varphi) = P_l^m(\cos\theta)\, e^{-\mathrm{i}m\varphi}, \tag{1}$$

where P_l^m is the associated Legendre polynomial, θ is the polar angle from the z-axis, and φ is the azimuthal angle. (We use the unnormalized convention; the characteristic equation is independent of the normalization choice.)

The key property we use is that spherical harmonics are *eigenfunctions of the Laplace–Beltrami operator* (the angular part of the Laplacian):

$$\nabla^2_{\text{angular}} Y_l^m = -\frac{l(l+1)}{r^2} Y_l^m. \tag{2}$$

This means that for any function of the form $f(r)\, Y_l^m(\theta, \varphi)$, the full scalar Laplacian is

$$\nabla^2[f(r)\, Y_l^m] = \left[f'' + \frac{2}{r} f' - \frac{l(l+1)}{r^2} f \right] Y_l^m, \tag{3}$$

where primes denote $\mathrm{d}/\mathrm{d}r$.

For axisymmetric problems we take $m = 0$, so $Y_l^0 \propto P_l(\cos\theta)$ is just the Legendre polynomial. Reid's analysis holds for all m; the characteristic equation turns out to be independent of m.

2.2 Spherical Bessel functions

The ordinary Bessel equation of order ν is

$$w'' + \frac{1}{z} w' + \left(1 - \frac{\nu^2}{z^2}\right) w = 0, \tag{4}$$

with solutions $J_\nu(z)$ (Bessel functions of the first kind) and $Y_\nu(z)$ (second kind, singular at $z = 0$).

The *spherical Bessel equation* of order l is

$$v'' + \frac{2}{z} v' + \left(1 - \frac{l(l+1)}{z^2}\right) v = 0, \tag{5}$$

and arises when separating the Helmholtz equation $(\nabla^2 + k^2)F = 0$ in spherical coordinates. Its regular solution is the *spherical Bessel function of the first kind*,

$$j_l(z) = \sqrt{\frac{\pi}{2z}}\, J_{l+1/2}(z). \tag{6}$$

Note

Any function $U(x) = x\, j_l(qx)$ satisfies the ODE

$$U'' - \frac{l(l+1)}{x^2} U + q^2 U = 0 \tag{7}$$

(the prime here is $\mathrm{d}/\mathrm{d}x$). This follows immediately by the substitution $U = xv$ with $x = qr/R$, which reduces (7) to the spherical Bessel equation (5) for v.

Proof. Let $U = xv(x)$. Then $U' = v + xv'$ and $U'' = 2v' + xv''$. Substituting:

$$2v' + xv'' - \frac{l(l+1)}{x}v + q^2xv = x\left[v'' + \frac{2}{x}v' + q^2v - \frac{l(l+1)}{x^2}v\right] = 0,$$

which is precisely (5) with $z = qx$. $\square$

We will also need the derivative identity. Using the standard recurrence $j_l'(z) = \frac{l}{z}j_l(z) - j_{l+1}(z)$, we obtain

$$\frac{q\, j_l'(q)}{j_l(q)} = l - q\,\frac{j_{l+1}(q)}{j_l(q)}. \tag{8}$$

Reid defines the ratio

$$Q_{l+1/2}(q) \equiv \frac{J_{l+3/2}(q)}{J_{l+1/2}(q)} = \frac{j_{l+1}(q)}{j_l(q)}, \tag{9}$$

so (8) becomes $qj_l'(q)/j_l(q) = l - qQ_{l+1/2}(q)$.

2.3 The poloidal decomposition

Any divergence-free vector field $\boldsymbol{u}$ can be uniquely decomposed into a *toroidal* part (involving only $\nabla \times (\Psi\hat{r})$, with no radial component) and a *poloidal* part (involving $\nabla \times \nabla \times (\Phi\hat{r})$, with non-zero radial component). For axisymmetric flow with no swirl, the toroidal part vanishes, and $\boldsymbol{u}$ is purely poloidal.

For a poloidal field with angular dependence Y_l^m, the entire velocity field is determined by its radial component u_r. Incompressibility then gives u_θ, and there are no other free components. This is why it suffices to write down a single scalar ODE for $U(x)$, where $u_r \propto U(x)/x^2$.

3 Problem Setup

3.1 Equilibrium state

A liquid drop of density ρ, viscosity $\mu = \rho\nu$, and surface tension T_1 is in equilibrium as a sphere of radius R. With the external pressure set to zero, the *Young–Laplace equation* gives the internal equilibrium pressure:

$$p_0 = \frac{2T_1}{R}. \tag{10}$$

Young–Laplace Equation

For an interface with principal radii of curvature R_1 and R_2, the pressure jump across the interface is $\Delta p = T_1(1/R_1 + 1/R_2)$. For a sphere, $R_1 = R_2 = R$, giving $\Delta p = 2T_1/R$.

3.2 Perturbed surface

We consider a small deformation of the spherical surface of the form

$$r = R[1 + \epsilon\, Y_l^m(\theta, \varphi)]\,, \qquad \epsilon \ll 1. \tag{11}$$

We look for solutions with time dependence

$$\epsilon(t) = \epsilon_0\, e^{-\sigma t}, \tag{12}$$

where σ is a complex number to be determined. The convention is chosen so that $\mathrm{Re}(\sigma) > 0$ corresponds to decay.

Note

Why $e^{-\sigma t}$ rather than $e^{+\mathrm{i}\omega t}$? Reid follows the convention of Chandrasekhar, where σ is complex and damped oscillations correspond to $\sigma = \gamma + \mathrm{i}\omega_d$ with $\gamma > 0$ (decay) and $\omega_d > 0$ (oscillation frequency). In the inviscid limit, $\gamma \to 0$ and $\sigma \to \pm\mathrm{i}\omega_d$.

3.3 Inviscid frequencies

For completeness, we record Lamb's result for the oscillation frequencies in the *absence* of viscosity. The frequency of a spherical harmonic mode of order l is

$$\sigma_{l;0}^2 = l(l-1)(l+2)\,\frac{T_1}{\rho R^3}. \tag{13}$$

These are real (undamped oscillations) and independent of the azimuthal order m. The $l = 2$ mode, for example, corresponds to the oblate–prolate oscillation familiar from a wobbling drop of water.

Physical check

Equation (13) is dimensionally consistent: $[T_1/(\rho R^3)] = [\mathrm{N/m}]/[\mathrm{kg/m^3} \cdot \mathrm{m^3}] = [\mathrm{kg/s^2}]/[\mathrm{kg}] = [\mathrm{s^{-2}}]$. The $l = 1$ mode gives $\sigma_{1;0}^2 = 0$, corresponding to a rigid translation (no restoring force).

4 Linearized Governing Equations

The departures from equilibrium are governed by the linearized incompressible Navier–Stokes equations,

$$\frac{\partial \boldsymbol{u}}{\partial t} = -\nabla \frac{\delta p}{\rho} - \nu\, \nabla \times \nabla \times \boldsymbol{u}, \qquad \nabla \cdot \boldsymbol{u} = 0. \tag{14}$$

Vector identity

For incompressible flow, $\nabla \cdot \boldsymbol{u} = 0$ implies $\nabla^2 \boldsymbol{u} = -\nabla \times \nabla \times \boldsymbol{u}$, so (14) is equivalent to $\partial_t \boldsymbol{u} = -\nabla(\delta p/\rho) + \nu \nabla^2 \boldsymbol{u}$. Both forms appear in the literature.

With the time dependence (12), all fields vary as $e^{-\sigma t}$, so $\partial_t \to -\sigma$. The momentum equation becomes

$$-\sigma \boldsymbol{u} = -\nabla \frac{\delta p}{\rho} + \nu \nabla^2 \boldsymbol{u}. \tag{15}$$

5 The Pressure Field

5.1 Pressure satisfies Laplace's equation

Derivation: $\nabla^2(\delta p/\rho) = 0$

Take the divergence of (15):

$$-\sigma \underbrace{(\nabla \cdot \boldsymbol{u})}_{=0} = -\nabla^2 \left(\frac{\delta p}{\rho} \right) + \nu \underbrace{\nabla \cdot (\nabla^2 \boldsymbol{u})}_{\nabla^2(\nabla \cdot \boldsymbol{u}) = 0},$$

where we used $\nabla \cdot \nabla^2 \boldsymbol{u} = \nabla^2(\nabla \cdot \boldsymbol{u}) = 0$. Therefore:

$$\nabla^2 \left(\frac{\delta p}{\rho} \right) = 0. \tag{16}$$

The pressure perturbation satisfies Laplace's equation. We need the solution that: (a) has angular dependence Y_l^m, and (b) is *regular at the origin* $r = 0$.

From (3), the radial ODE for $f(r)$ in $\nabla^2(f\,Y_l^m) = 0$ is

$$f'' + \frac{2}{r} f' - \frac{l(l+1)}{r^2} f = 0. \tag{17}$$

The general solution is $f(r) = Ar^l + Br^{-(l+1)}$. Regularity at $r = 0$ requires $B = 0$. Writing $x = r/R$, we therefore have

$$\frac{\delta p}{\rho} = \epsilon\, P_0\, x^l\, Y_l^m, \tag{18}$$

where P_0 is a constant of integration (with dimensions of velocity squared) to be determined from the boundary conditions.

5.2 Scalar potential for the pressure gradient

Since $\nabla(\delta p/\rho)$ is a purely poloidal, divergence-free vector field with angular structure Y_l^m, it can be represented through a single scalar function $\Pi(x)$ via its radial component:

$$\left(\nabla \frac{\delta p}{\rho}\right)_r = \epsilon_0\, \sigma^2 R\, \frac{\Pi(x)}{x^2}\, Y_l^m\, e^{-\sigma t}. \tag{19}$$

Computing the radial component of $\nabla(\epsilon P_0 x^l Y_l^m)$:

$$\left(\nabla \frac{\delta p}{\rho}\right)_r = \frac{\partial}{\partial r}\big(\epsilon P_0 x^l Y_l^m\big) = \epsilon P_0\, \frac{l\, x^{l-1}}{R}\, Y_l^m = \epsilon_0 e^{-\sigma t} P_0\, \frac{l}{R}\, x^{l-1}\, Y_l^m.$$

Comparing with (19), we need $\epsilon_0 \sigma^2 R\, \Pi(x)/x^2 = \epsilon_0 P_0 l x^{l-1}/R$, which gives $\Pi(x) = \Pi_0\, x^{l+1}$ with

$$\Pi(x) = \Pi_0\, x^{l+1}, \qquad \Pi_0 = \frac{l}{\sigma^2 R^2}\, P_0. \tag{20}$$

6 The Velocity Field and its Governing ODE

6.1 Poloidal representation

Since the pressure gradient is purely poloidal, the momentum equation (15) forces $\boldsymbol{u}$ to also be purely poloidal (the toroidal part would have no driver). For a poloidal field with angular dependence Y_l^m, we write the radial component as

$$u_r = \epsilon_0\, \sigma R\, \frac{U(x)}{x^2}\, Y_l^m\, e^{-\sigma t}, \tag{21}$$

where $U(x)$ is a dimensionless scalar function.

Why $U(x)/x^2$?

The scaling $1/x^2$ is conventional and simplifies the algebra. It arises from the stream-function representation of poloidal flow, and ensures the ODE for U takes the standard form (24) below.

6.2 Derivation of the ODE for $U(x)$

Full derivation

We substitute the ansatz $u_r = \sigma R\, G(x)\, Y_l^m$ (where $G = U/x^2$, factoring out the $\epsilon_0 e^{-\sigma t}$ terms throughout) into the r-component of (15). The r-component of the vector Laplacian for an axisymmetric poloidal field is:

$$\left[\nabla^2 \boldsymbol{u}\right]_r = \nabla^2 u_r + \frac{2}{r^2} u_r + \frac{2}{r}\frac{\partial u_r}{\partial r}. \tag{22}$$

Derivation of (22). The vector Laplacian r-component in spherical coordinates is:

$$\left[\nabla^2 \boldsymbol{u}\right]_r = \nabla^2 u_r - \frac{2u_r}{r^2} - \frac{2}{r^2 \sin\theta}\frac{\partial(\sin\theta\, u_\theta)}{\partial\theta}.$$

From incompressibility, $(1/r\sin\theta)\,\partial(\sin\theta\, u_\theta)/\partial\theta = -(1/r^2)\,\partial(r^2 u_r)/\partial r$, so:

$$\frac{1}{\sin\theta}\frac{\partial(\sin\theta\, u_\theta)}{\partial\theta} = -\frac{r}{r^2}\,\partial_r(r^2 u_r) = -(2u_r + r u_r').$$

Substituting:

$$\left[\nabla^2 \boldsymbol{u}\right]_r = \nabla^2 u_r - \frac{2u_r}{r^2} + \frac{2}{r^2}(2u_r + r u_r') = \nabla^2 u_r + \frac{2u_r}{r^2} + \frac{2}{r}u_r'.$$

Now, for $u_r = \sigma R\, G(x)\, Y_l^m$ with $G = G(x)$ and $x = r/R$:

$$\nabla^2 u_r = \sigma R\, Y_l^m \left[G'' + \frac{2}{x}G' - \frac{l(l+1)}{x^2}G\right]\frac{1}{R^2} \quad \text{(from (3))},$$

$$\frac{2}{r}u_r' = \frac{2}{Rx}\cdot \sigma\, G'\, Y_l^m = \frac{2\sigma}{Rx}\, G'\, Y_l^m,$$

$$\frac{2u_r}{r^2} = \frac{2\sigma R G}{R^2 x^2}\, Y_l^m = \frac{2\sigma G}{R x^2}\, Y_l^m.$$

Combining, the r-component of the momentum equation $-\sigma\boldsymbol{u} = -\nabla(\delta p/\rho) + \nu\nabla^2\boldsymbol{u}$ becomes (after canceling $Y_l^m e^{-\sigma t}$ throughout and using $x = r/R$):

$$-\sigma^2 R\, G = -\frac{P_0 l\, x^{l-1}}{R} + \frac{\nu\sigma}{R}\left[G'' + \frac{4}{x}G' + \frac{2 - l(l+1)}{x^2}G\right]. \tag{23}$$

Multiplying through by $R/(\nu\sigma)$ and using $q^2 = \sigma R^2/\nu$:

$$-q^2 G = -\frac{P_0 l x^{l-1}}{\sigma\nu} + G'' + \frac{4}{x}G' + \frac{2 - l(l+1)}{x^2}G.$$

Now substitute $G = U/x^2$. One computes:

$$G'' + \frac{4}{x}G' + \frac{2 - l(l+1)}{x^2}G = \frac{U'' - l(l+1)U/x^2}{x^2}.$$

(This identity is most easily verified by direct substitution, expanding each term.) Multiplying through by x^2:

$$U'' - \frac{l(l+1)}{x^2}U + q^2 U = \frac{P_0 l x^{l+1}}{\sigma\nu}.$$

Using (20), $P_0 = \sigma^2 R^2 \Pi_0 / l$, so $P_0 l/(\sigma\nu) = \sigma R^2 \Pi_0/\nu = q^2 \Pi_0$. Therefore:

$$\boxed{\left[\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{l(l+1)}{x^2} + q^2\right] U(x) = q^2 \Pi_0\, x^{l+1},} \tag{24}$$

where

$$q^2 = \frac{\sigma R^2}{\nu}. \tag{25}$$

This is Reid's Eq. (9). □

6.3 Solving the ODE

Equation (24) is a second-order linear inhomogeneous ODE. We find the general solution by superposing a particular solution and the general homogeneous solution.

Particular solution. Try $U_p = \Pi_0 \, x^{l+1}$:

$$U_p'' = l(l+1)\Pi_0 \, x^{l-1}, \quad \frac{l(l+1)}{x^2} U_p = l(l+1)\Pi_0 \, x^{l-1}, \quad q^2 U_p = q^2 \Pi_0 \, x^{l+1}.$$

The first two terms cancel exactly, leaving $q^2 \Pi_0 \, x^{l+1}$ on the left, matching the right-hand side. So

$$U_p(x) = \Pi_0 \, x^{l+1}. \tag{26}$$

Homogeneous solution. The homogeneous equation $U'' - l(l+1)U/x^2 + q^2 U = 0$ has solutions $x \, j_l(qx)$ and $x \, n_l(qx)$, where j_l and n_l are spherical Bessel functions of the first and second kind (see Section 2.2). Since $n_l(z) \sim -(2l-1)!! \, z^{-(l+1)}$ as $z \to 0$ (diverges at the origin), regularity requires we discard n_l. The regular homogeneous solution is thus $C \, x \, j_l(qx)$.

General solution.

$$U(x) = C \, x \, j_l(qx) + \Pi_0 \, x^{l+1}, \tag{27}$$

where C and Π_0 are determined by the boundary conditions.

7 Boundary Conditions

Three boundary conditions must be imposed at the drop surface $x = 1$.

7.1 BC1: Kinematic condition

The radial velocity of the fluid at the surface must equal the velocity at which the surface itself moves.

Derivation of BC1

The surface is at $r = R[1 + \epsilon \, Y_l^m]$ with $\epsilon = \epsilon_0 e^{-\sigma t}$. The radial velocity of a surface element is:

$$\left. \frac{\partial r}{\partial t} \right|_{\text{surface}} = -\sigma R \, \epsilon_0 e^{-\sigma t} \, Y_l^m.$$

The fluid radial velocity at $x = 1$ (from (21)) is:

$$u_r \big|_{x=1} = \epsilon_0 \, \sigma R \, U(1) \, Y_l^m \, e^{-\sigma t}.$$

Setting these equal:

$$\epsilon_0 \, \sigma R \, U(1) = -\sigma R \, \epsilon_0.$$

Canceling $\epsilon_0 \sigma R$:

$$U(1) = -1. \tag{28}$$

The kinematic condition is imposed at $x = 1$ (the unperturbed surface) because evaluating at the actual surface $r = R[1 + \epsilon Y_l^m]$ introduces only $O(\epsilon^2)$ corrections.

Note

The sign convention $U(1) = -1$ follows from the positive-σ prefactor in (21) with $\epsilon_0 > 0$ encoding the outward displacement. This matches Reid (1960): his Eq. (8) uses the same positive-σR prefactor and his Eq. (11) states $U(1) = -1$.

7.2 BC2: Tangential stress condition

At a free surface with no viscous exterior fluid, the tangential viscous stress must vanish.

The $r\theta$-component of the viscous stress tensor in spherical coordinates is:

$$\tau_{r\theta} = \mu \left[r \frac{\partial}{\partial r}\left(\frac{u_\theta}{r}\right) + \frac{1}{r}\frac{\partial u_r}{\partial \theta} \right].$$

For a poloidal velocity field with harmonic structure Y_l^m, using the stream-function representation and the incompressibility constraint to express u_θ in terms of u_r, one can show that $\tau_{r\theta} = 0$ at $x = 1$ reduces to:

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{2}{x}\frac{\mathrm{d}}{\mathrm{d}x} + \frac{l(l+1)}{x^2} \right] U = 0 \qquad \text{at } x = 1. \tag{29}$$

Evaluating BC2 on the general solution

For the particular solution $U_p = \Pi_0 x^{l+1}$:

$$\mathcal{L}_2[x^{l+1}] = l(l+1)x^{l-1} - 2(l+1)x^{l-1} + l(l+1)x^{l-1} = 2(l-1)(l+1)x^{l-1}.$$

At $x = 1$: $\mathcal{L}_2[U_p]|_{x=1} = 2(l-1)(l+1)\Pi_0$.
For the homogeneous part $U_h = Cxj_l(qx)$, using $U_h'' = -q^2 x j_l + l(l+1)j_l/x$ (from the Bessel equation) and $U_h' = j_l + qxj_l'$:

$$\mathcal{L}_2[U_h]\big|_{x=1} = C\left[-q^2 j_l(q) + 2\big(l(l+1) - 1\big)j_l(q) - 2qj_l'(q)\right].$$

Setting the sum to zero gives one equation relating C and Π_0.

7.3 BC3: Normal stress condition

This boundary condition couples pressure, viscous stress, and surface curvature through surface tension.

7.3.1 Mean curvature of the deformed surface

Curvature of the perturbed sphere

For the surface (11), the principal curvatures can be computed by standard differential geometry. To first order in ϵ:

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R}\left[\, 2 + (l-1)(l+2)\, \epsilon\, Y_l^m \,\right]. \tag{30}$$

Physical interpretation: The $2/R$ term is the equilibrium curvature. The perturbation contributes $(l-1)(l+2)\epsilon Y_l^m/R$. The factor $(l-1)(l+2)$ is the same one that appears in the inviscid frequency (13). Both arise from the curvature response to a harmonic deformation.

7.3.2 Normal stress balance

The radial normal stress inside the drop is:

$$-p_{rr} = p + \delta p - 2\mu \frac{\partial u_r}{\partial r}. \tag{31}$$

The condition at the free surface is:

$$-p_{rr} = T_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \tag{32}$$

Derivation of BC3

At $O(\epsilon^0)$: the equilibrium Young–Laplace condition $p_0 = 2T_1/R$ is automatically satisfied (the drop is in equilibrium). At $O(\epsilon^1)$:

$$\delta p - 2\mu \frac{\partial u_r}{\partial r}\bigg|_{r=R} = T_1\, \frac{(l-1)(l+2)}{R}\, \epsilon\, Y_l^m.$$

Using (18) with $x = 1$: $\delta p|_{x=1} = \rho \epsilon P_0 Y_l^m$ (since $\delta p/\rho = \epsilon P_0 x^l Y_l^m$ at $x = 1$).
For the viscous term, with $u_r = \epsilon_0 \sigma R\, G(x) Y_l^m e^{-\sigma t}$ and $G = U/x^2$:

$$2\mu \frac{\partial u_r}{\partial r}\bigg|_{x=1} = 2\rho\nu \cdot \sigma \cdot \epsilon_0 e^{-\sigma t}\, Y_l^m\; \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{U}{x^2}\right)\bigg|_{x=1} = 2\rho\nu\sigma\, \epsilon\, Y_l^m \left[\, U'(1) - 2U(1) \,\right].$$

Dividing through by $\rho\epsilon\, Y_l^m$ and using $\mu = \rho\nu$:

$$(l-1)(l+2)\frac{T_1}{\rho R} = P_0 - 2\nu\sigma\left[\, U'(1) - 2U(1) \,\right]. \tag{33}$$

Note

BC3 is the only equation containing T_1 explicitly. Reid's key move (next section) is to rescale parameters so that T_1 drops out entirely.

8 The Characteristic Equation

8.1 Eliminating T_1 via rescaling

Define the dimensionless ratio

$$\alpha^2 \equiv \frac{\sigma_{l;0}\, R^2}{\nu}, \tag{34}$$

which measures the inviscid frequency in units of the viscous diffusion rate ν/R^2. Equivalently, $\alpha = \mathrm{Oh}^{-1/2} \cdot [l(l-1)(l+2)]^{1/4}$ in terms of the Ohnesorge number.

Eliminating T_1

From (13): $\sigma_{l;0}^2 = l(l-1)(l+2)T_1/(\rho R^3)$, so $(l-1)(l+2)T_1/(\rho R) = \sigma_{l;0}^2 R^2/l$. Using (20), $P_0 = \sigma^2 R^2 \Pi_0/l$. Substituting into BC3 (33):

$$\frac{\sigma_{l;0}^2 R^2}{l} = \frac{\sigma^2 R^2 \Pi_0}{l} - 2\nu\sigma[U'(1) - 2U(1)]\,.$$

Multiply through by lR^2/ν^2:

$$\frac{\sigma_{l;0}^2 R^4}{\nu^2} = \frac{\sigma^2 R^4 \Pi_0}{\nu^2} - \frac{2l\sigma R^2}{\nu}[U'(1) - 2U(1)]$$
$$\alpha^4 = q^4 \Pi_0 - 2l\, q^2 [U'(1) - 2U(1)]\,.$$

This can be written compactly as

$$\alpha^4 = q^2 \big\{\, q^2 \Pi_0 - 2l\big[\, U'(1) - 2U(1)\,\big]\big\}\,. \tag{35}$$

T_1 and ρ have cancelled. The problem in terms of (α, q) is universal.

8.2 Applying BCs to the general solution

We now apply BC1 (28) and BC2 (29) to the general solution (27) to express C and Π_0 in terms of Bessel functions, then substitute into (35) to obtain the characteristic equation.

Applying BC1.

$$U(1) = C\, j_l(q) + \Pi_0 = -1. \tag{36}$$

Applying BC2. Using the results from the note in Section 6.2:

$$C\left[-q^2 j_l + 2(l^2+l-1)j_l - 2qj_l'\right] + 2(l^2-1)\Pi_0 = 0. \tag{37}$$

Solving for C and Π_0. From (36): $\Pi_0 = -1 - Cj_l(q)$. Substituting into (37) and simplifying:

$$C\left[(2l-q^2)j_l - 2qj_l'\right] = 2(l^2-1).$$

Using $qj_l'/j_l = l - qQ_{l+1/2}$ from (8) and the definition (9):

$$C = \frac{2(l-1)(l+1)}{(2l-q^2)j_l(q) - 2q\,j_l'(q)}. \tag{38}$$

The expression for Π_0 follows from (36).

Substituting into (35). After substituting the expressions for C, Π_0, $U'(1)$, and $U(1)$ into (35), and expressing all Bessel function ratios in terms of $Q_{l+1/2}(q) = j_{l+1}(q)/j_l(q)$ via the recurrence (8), one obtains the characteristic equation:

The Characteristic Equation (Reid 1960, Eq. 19)

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q^2}\left[l + (l+1)\,\frac{q - 2l\,Q_{l+1/2}(q)}{q - 2Q_{l+1/2}(q)}\right], \tag{39}$$

where $q^2 = \sigma R^2/\nu$, $\alpha^2 = \sigma_{l;0}R^2/\nu$, and $Q_{l+1/2}(q) = J_{l+3/2}(q)/J_{l+1/2}(q) = j_{l+1}(q)/j_l(q)$.

Universality

This is identical to Chandrasekhar's characteristic equation for a viscous *self-gravitating* liquid globe, with the identification that the self-gravitational parameter maps to α^2. The physical restoring force (surface tension vs. self-gravity) enters only through $\sigma_{l;0}$, which defines α. (39) holds for any spherically symmetric restoring force.

Algebraic derivation remark

The algebraic steps from (38) to (39) involve expressing $q^2\Pi_0 - 2l[U'(1) - 2U(1)]$ entirely in terms of Bessel ratios. Key simplifications use the identity

$$j_{l+1}(q) = (2l+1)j_l(q)/q - j_{l-1}(q)$$

(spherical Bessel recurrence) and the relation (8). The algebra is lengthy; each step uses only linear algebra and these Bessel identities.

9 Structure of the Solutions

9.1 Roots of the characteristic equation

Equation (39) is a transcendental equation for $q^2 = \sigma R^2/\nu$ (or equivalently for σ), given the physical parameters encoded in α. It has infinitely many solutions.

Physical picture. The Bessel function $J_{l+1/2}(q)$ in the denominator of $Q_{l+1/2}$ has infinitely many real zeros $q_1 < q_2 < \cdots$ on the positive real axis. Near each zero the characteristic equation (39) has a pole, and each pole "captures" one pair of roots. These correspond to increasingly high radial overtones of the oscillation, i.e., modes with more and more nodal surfaces in the radial direction inside the drop.

Ordering by decay rate. Roots with *smaller* real part of σ (i.e., slower decay) are the physically dominant ones: they are the last to die out and govern the observable oscillation of the drop. The two roots with the smallest $\mathrm{Re}(\sigma)$ correspond to the fundamental surface oscillation at harmonic order l; all higher roots decay exponentially faster.

Sign convention caution

In Reid's convention, $\epsilon \sim e^{-\sigma t}$ with $\mathrm{Re}(\sigma) > 0$ meaning decay. So the *smallest positive* real part corresponds to the *slowest* decay, i.e., the dominant mode. This is the opposite of the situation in stability analysis where one looks for the most-positive growth rate.

9.2 Two regimes: oscillatory vs. aperiodic

For fixed l, the character of the two dominant roots depends on α^2:

- **Large α^2 (low viscosity, large drop):** the two dominant roots are complex conjugates, $\sigma = \gamma \pm \mathrm{i}\omega_d$, corresponding to *damped oscillations*.
- **Small α^2 (high viscosity, small drop):** the two dominant roots are both real and positive, corresponding to two *aperiodic decaying modes* (overdamped).

The transition occurs at a critical value of α^2. For $l = 2$, this critical point was determined numerically by Chandrasekhar:

$$\sigma_{2;0} R^2/\nu = 3.69, \qquad \sigma_{2;\nu}/\sigma_{2;0} = 0.968. \tag{40}$$

For water in air ($T_1 = 74$ dyn/cm, $\nu = 0.01$ cm^2/s), this gives a critical radius $R_c \approx$ 0.23 mm.

- Drops with $R > R_c$: oscillate with damping.
- Drops with $R < R_c$: aperiodically damped (no oscillations).

9.3 The small-viscosity limit

As $\nu \to 0$ (or $q \to \infty$), the spherical Bessel functions have the large-argument asymptotics $J_\nu(q) \sim \sqrt{2/(\pi q)}\cos(q - \nu\pi/2 - \pi/4)$, which for $\nu = l + \frac{1}{2}$ reduces to $J_{l+1/2}(q) \sim \sqrt{2/(\pi q)}\sin(q - l\pi/2)$. The ratio $Q_{l+1/2}(q) = j_{l+1}(q)/j_l(q)$ therefore oscillates as $q \to \infty$ (with poles at every zero of j_l), but $Q_{l+1/2}(q)/q \to 0$ between poles, since the amplitude of $J_{l+3/2}/J_{l+1/2}$ remains $O(1)$ away from zeros. The bracket on the right-hand side of (39), $l + (l+1)(q - 2l\,Q_{l+1/2})/(q - 2Q_{l+1/2})$, approaches $l + (l+1)\cdot 1 = 2l+1$ as $q \to \infty$. The characteristic equation (39) then simplifies to:

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)(2l+1)}{q^2},$$

which is a quadratic in q^2/α^2, yielding

$$\sigma_{l;\nu} = (l-1)(2l+1)\,\frac{\nu}{R^2} \pm \mathrm{i}\,\sigma_{l;0}, \tag{41}$$

recovering *Lamb's classical result*: viscosity adds a real damping rate $(l-1)(2l+1)\nu/R^2$ to the inviscid oscillation frequency $\pm\mathrm{i}\sigma_{l;0}$.

10 Connection to Molaček & Bush (2012)

In their quasi-static model of drop impact, Molaček & Bush parameterize the kinetic energy and viscous dissipation associated with each surface harmonic mode m (their notation) via two coefficients $A_m(\mathrm{Oh}_m)$ and $D_m(\mathrm{Oh}_m)$, which appear in (their Eq. 31):

$$\text{K.E.} = \pi\rho R_0^5 \dot{B}^2 \sum_m A_m \frac{2b_m^2}{m(2m+1)}, \qquad D = 8\pi\mu R_0^3 \dot{B}^2 \sum_m D_m \frac{m}{2m+1} b_m^2.$$

These coefficients are defined so that the two roots of the quadratic

$$A_m b^2 - 2aD_m b + 1 = 0, \qquad a \equiv \mathrm{Oh}\sqrt{m(m-1)(m+2)}, \tag{42}$$

reproduce the two dominant roots of Reid's characteristic equation (39) for harmonic order $l = m$. The variable identification connecting MB's quadratic to Reid's transcendental equation is $q^2 = (b/a)\cdot\alpha^2$ (i.e. $b = a\cdot q^2/\alpha^2$), under which MB's $W(b/a) = Q_{l+1/2}(q)/q$. In other words, A_m and D_m encode the *result* of solving Reid's problem at each order m, compressed into two numbers per mode via a quadratic fit to the dominant eigenvalue pair.

Workflow.

1. For each mode m and Ohnesorge number Oh, evaluate $\alpha^2 = \mathrm{Oh}^{-1}\sqrt{m(m-1)(m+2)}$ and $q^2 = \sigma R^2/\nu$.

2. Numerically solve (39) for the two roots $b_{1,2}$ with the smallest $\mathrm{Re}(\sigma)$.

3. Read off A_m and D_m by matching the quadratic: sum of roots $= 2aD_m/A_m$; product of roots $= 1/A_m$.

4. Tabulate or fit $A_m(\mathrm{Oh}_m)$ and $D_m(\mathrm{Oh}_m)$ for use in the Lagrangian equation of motion.

Justification for discarding higher roots. The quasi-static assumption restricts the drop shape to evolve through a one-parameter family of equilibrium shapes. Within that family, only the fundamental surface mode at each harmonic order l is representable. Higher radial overtones have internal nodal surfaces absent from the quasi-static shape family. They decay on timescales $\ll (R^2/\nu)$, well separated from the impact timescale.

11 Summary of Key Equations

Equation	Physical content
$\sigma_{l;0}^2 = l(l-1)(l+2)\,T_1/(\rho R^3)$	Inviscid capillary frequency
$q^2 = \sigma R^2/\nu$	Complex decay rate, scaled
$\alpha^2 = \sigma_{l;0} R^2/\nu$	Inviscid frequency, scaled
$Q_{l+1/2}(q) = J_{l+3/2}(q)/J_{l+1/2}(q)$	Bessel function ratio
$\alpha^4/q^4 + 1 = (2(l-1)/q^2)[\cdots]$	Reid's characteristic equation
$\sigma = (l-1)(2l+1)\nu/R^2 \pm \mathrm{i}\,\sigma_{l;0}$	Small-ν (Lamb) limit

12 Extension to Viscoelastic Drops: the Oldroyd-B Model

Every result in Sections 1–9 assumes the viscous stress is instantaneously proportional to the rate of strain, $\boldsymbol{\tau} = 2\mu\,\mathbf{e}$. Real polymer solutions violate this. A drop of polyethylene oxide (PEO) in water, a silicone oil of moderate molecular weight, or a biological fluid such as mucus all retain a *memory* of past deformations. Stress does not vanish the instant deformation stops; it relaxes on a characteristic timescale set by the polymer microstructure.

The **Oldroyd-B model** is the simplest constitutive law that captures this memory and remains analytically tractable. It is the canonical model for dilute polymer solutions: a Newtonian solvent (viscosity μ_s) in which elastic polymer dumbbells are suspended (contributing additional viscosity μ_p, with $\mu = \mu_s + \mu_p$ the total). The model has two timescales: a *relaxation time* λ_1 (how long polymer stress persists after deformation stops) and a *retardation time* $\lambda_2 < \lambda_1$ (how long the combined fluid takes to respond to a new deformation). Setting $\lambda_2 = 0$ reduces Oldroyd-B to the Maxwell model; setting $\lambda_1 = \lambda_2$ recovers a Newtonian fluid.

Reid's characteristic equation (39) survives generalisation to Oldroyd-B. The left-hand side α^4/q^4+1 is unchanged. The right-hand side is Reid's right-hand side with q replaced by a modified wavenumber q_*, a rational function of q^2. This rational structure introduces a new type of pole, the *retardation pole*, which captures additional eigenvalues beyond Reid's fundamental pair and eventually causes the two-number $\{A_l, D_l\}$ picture to break down.

12.1 The Oldroyd-B constitutive law

12.1.1 Physical motivation: the mechanical analog

Spring-dashpot picture of Oldroyd-B

Imagine the fluid as two *parallel* stress-carrying branches:

- **Solvent branch:** a pure dashpot of viscosity μ_s. Stress responds instantaneously to strain rate: $\boldsymbol{\tau}_s = 2\mu_s\,\mathbf{e}$. No memory.
- **Polymer branch:** a Hookean spring (elastic modulus $G = \mu_p/\lambda_1$) in *series* with a dashpot of viscosity μ_p. The spring stores elastic energy when the dumbbell is stretched; the dashpot relaxes that energy on timescale $\lambda_1 = \mu_p/G$.

The total stress is $\boldsymbol{\tau} = \boldsymbol{\tau}_s + \boldsymbol{\tau}_p$. The retardation time

$$\lambda_2 = \lambda_1\,\frac{\mu_s}{\mu_s+\mu_p} = \beta_s\,\lambda_1 \tag{43}$$

is the timescale on which the *combined* system (solvent + polymer) first responds to a new deformation: the solvent wants to respond immediately, but the spring in the polymer arm initially resists, delaying the total response.

12.1.2 The constitutive equation: two equivalent forms

In terms of the two raw timescales (λ_1, λ_2):

$$\boldsymbol{\tau} + \lambda_1\,\overset{\nabla}{\boldsymbol{\tau}} = 2\mu\Big(\mathbf{e} + \lambda_2\,\overset{\nabla}{\mathbf{e}}\Big), \qquad 0 \le \lambda_2 < \lambda_1, \tag{44}$$

where $\overset{\nabla}{(\cdot)}$ is the upper-convected time derivative:

$$\overset{\nabla}{\mathbf{A}} = \frac{\partial \mathbf{A}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\mathbf{A} - (\nabla\boldsymbol{u})\cdot\mathbf{A} - \mathbf{A}\cdot(\nabla\boldsymbol{u})^{\top}. \tag{45}$$

In terms of the retardation ratio $\beta_s \equiv \lambda_2/\lambda_1$:

$$\boldsymbol{\tau} + \lambda_1\,\overset{\nabla}{\boldsymbol{\tau}} = 2\mu\Big(\mathbf{e} + \beta_s\lambda_1\,\overset{\nabla}{\mathbf{e}}\Big), \qquad \beta_s \in [0,1), \tag{46}$$

with $\beta_s = \mu_s/(\mu_s + \mu_p)$ the solvent viscosity fraction. The two forms (44) and (46) are identical; we carry both throughout because each makes different limits transparent.

Upper-convected derivative at linearised order

All fields are $O(\epsilon)$. In the upper-convected derivative (45), the terms $(\boldsymbol{u} \cdot \nabla)\mathbf{A}$, $(\nabla \boldsymbol{u}) \cdot \mathbf{A}$, and $\mathbf{A} \cdot (\nabla \boldsymbol{u})^\top$ are each products of two $O(\epsilon)$ quantities, hence $O(\epsilon^2)$. At linear order:

$$\overset{\triangledown}{\boldsymbol{\tau}} \longrightarrow \frac{\partial \boldsymbol{\tau}}{\partial t} + O(\epsilon^2), \qquad \overset{\triangledown}{\mathbf{e}} \longrightarrow \frac{\partial \mathbf{e}}{\partial t} + O(\epsilon^2).$$

We record the full upper-convected form for completeness and frame- indifference, but drop the $O(\epsilon^2)$ terms immediately.

12.1.3 Parameter limits

Limit	Condition	Model	β_s
Maxwell	$\lambda_2 = 0$	Single-relaxation memory	0
Newtonian	$\lambda_1 = \lambda_2$	Viscous, viscosity μ	1
Solvent	$\lambda_1 \to 0$ at fixed $G = \mu_p/\lambda_1$, μ_s	Newtonian, viscosity μ_s	$\to 1$
Elastic solid	$\lambda_1 \to \infty$, $\beta_s = 0$, $G = \mu_p/\lambda_1$ fixed ($\mu_p \to \infty$)	Hookean solid, modulus G	0

12.2 Linearisation and effective viscosity

The linearisation strategy in this section follows Zrnić & Brenn [10], whose Eq. (3.3) establishes the same effective-viscosity replacement $\mu \to \mu_{\text{eff}}(\sigma)$ for the free-surface Oldroyd-B problem.

Linearised Oldroyd-B in the frequency domain

With time dependence $e^{-\sigma t}$ (our convention: $\text{Re}(\sigma) > 0$ for decay), $\partial_t \to -\sigma$. Dropping $O(\epsilon^2)$ upper-convected terms:

$$(1 - \lambda_1 \sigma)\, \boldsymbol{\tau} = 2\mu\, (1 - \lambda_2 \sigma)\, \mathbf{e}.$$

Dividing through by $(1 - \lambda_1 \sigma)$ (valid generically, since $\sigma = 1/\lambda_1$ is a special value examined separately):

$$\boldsymbol{\tau} = 2\mu_{\text{eff}}(\sigma)\, \mathbf{e}, \qquad \mu_{\text{eff}}(\sigma) = \mu\, \frac{1 - \lambda_2 \sigma}{1 - \lambda_1 \sigma}. \tag{47}$$

In both parameterisations simultaneously:

$$\mu_{\text{eff}}(\sigma) = \mu\, \frac{1 - \lambda_2 \sigma}{1 - \lambda_1 \sigma} = \mu\, \frac{1 - \beta_s \lambda_1 \sigma}{1 - \lambda_1 \sigma}. \tag{48}$$

The effective kinematic viscosity is $\nu_{\text{eff}} = \mu_{\text{eff}}/\rho$.

Analytic structure of $\mu_{\text{eff}}(\sigma)$

As a function of σ, μ_{eff} is a ratio of two linear factors:

- **Zero** at $\sigma = 1/\lambda_2$: solvent and polymer stresses cancel in the tangential direction. ($\lambda_2 = 0$ pushes this zero to infinity (the Maxwell model has no zero.)
- **Pole** at $\sigma = 1/\lambda_1$: polymer stress cannot relax; effective viscosity diverges (solid-like response).

Setting $\lambda_2 = 0$: $\mu_{\text{eff}} = \mu/(1 - \lambda_1\sigma)$, recovering the Maxwell effective viscosity. Setting $\lambda_1 = \lambda_2$: $\mu_{\text{eff}} = \mu$, recovering Newtonian. The Oldroyd-B model interpolates between these two limits.

12.2.1 The two Deborah numbers

Referenced to the inviscid oscillation frequency $\sigma_{l;0}$:

$$De_1 \equiv \lambda_1\,\sigma_{l;0}, \qquad De_2 \equiv \lambda_2\,\sigma_{l;0} = \beta_s\,De_1. \tag{49}$$

De_j is the ratio of the fluid's jth memory timescale to the drop's natural oscillation period.

Convention note

Zrnić & Brenn [10] define their Deborah numbers referenced to the capillary timescale $(T_1/(\rho R^3))^{1/2}$, whereas here they are referenced to $\sigma_{l;0} = [l(l-1)(l+2)]^{1/2}(T_1/(\rho R^3))^{1/2}$. Consequently $De_j^{\text{here}} = \sqrt{l(l-1)(l+2)}\,De_j^{\text{Z\&B}}$; numerical values are mode-dependent and should not be compared directly.

When $De_1 \ll 1$, the polymer relaxes many times per cycle and the drop behaves as Newtonian. When $De_1 \sim 1$, elasticity and surface tension compete on equal footing.

Expressing $\lambda_j\sigma$ in terms of q^2 and α^2

Recall $\sigma = q^2\nu/R^2$ and $\sigma_{l;0} = \alpha^2\nu/R^2$. Therefore:

$$\lambda_j\sigma = \lambda_j \cdot \frac{q^2\nu}{R^2} = \frac{\lambda_j\sigma_{l;0}}{\alpha^2\nu/R^2} \cdot \frac{q^2\nu}{R^2} = \frac{De_j \cdot q^2}{\alpha^2}. \tag{50}$$

Substituting into (47):

$$\mu_{\text{eff}} = \mu\,\frac{\alpha^2 - De_2\,q^2}{\alpha^2 - De_1\,q^2}, \qquad \nu_{\text{eff}} = \nu\,\frac{\alpha^2 - De_2\,q^2}{\alpha^2 - De_1\,q^2}. \tag{51}$$

All dimensional quantities have cancelled. The effective viscosity is a function of q^2/α^2 only.

12.3 Boundary condition audit

We have a free-surface drop in air. We examine each of the three boundary conditions.

12.3.1 BC1 (kinematic): unchanged

The kinematic condition equates the fluid radial velocity at $x = 1$ to the rate of change of the surface position. It is a purely geometric statement involving only the *velocity field*, not the stress. The Oldroyd-B constitutive law does not alter the velocity field structure. Therefore:

$$U(1) = -1. \qquad \text{(BC1, unchanged)}$$

12.3.2 BC2 (tangential stress): unchanged in form

The free-surface condition $\tau_{r\theta}|_{x=1} = 0$ becomes:

$$\mu_{\text{eff}}(\sigma)\left[r\frac{\partial}{\partial r}\left(\frac{u_\theta}{r}\right) + \frac{1}{r}\frac{\partial u_r}{\partial \theta}\right]_{r=R} = 0.$$

Since $\mu_{\text{eff}} \neq 0$ generically, we divide through. The condition reduces to exactly the same operator equation on U:

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{2}{x}\frac{\mathrm{d}}{\mathrm{d}x} + \frac{l(l+1)}{x^2}\right]U\bigg|_{x=1} = 0. \qquad \text{(BC2, unchanged)}$$

The zero of μ_{eff} at $\sigma = 1/\lambda_2$

At $\sigma = 1/\lambda_2$, the effective viscosity $\mu_{\text{eff}} \to 0$ and BC2 becomes $0 = 0$ for any U: it imposes no constraint. The frequency $\sigma = 1/\lambda_2$ is the *retardation pole* of q_*^2: as $w \to w_{\text{ret}} = \alpha^2/De_2$, $q_*^2 \to \infty$. It is not itself a root of the characteristic equation $F(w) = 0$. The actual retardation roots instead lie in a neighbourhood of w_{ret} (see Section 12.6). At those roots BC2 must be handled by a limiting argument. In practice these roots decay at rate $1/\lambda_2$, which is faster than the capillary timescale when $De_2 < 1$, and they are discarded as overtones. For $De_2 > 1$ a more careful treatment is required.

12.3.3 BC3 (normal stress): viscoelastic correction check

First normal stress difference at $O(\epsilon)$ for Oldroyd-B

For the Oldroyd-B family, the first normal stress difference is:

$$N_1 = \tau_{rr} - \tau_{\theta\theta}.$$

At the linearised level, $\boldsymbol{\tau} = 2\mu_{\text{eff}}\,\mathbf{e}$ with $\mathbf{e} = O(\epsilon)$, so $N_1 = O(\epsilon)$. Does this $O(\epsilon)$ quantity add a new term to BC3?

The normal stress balance at a free surface equates the jump in *radial* normal stress to the Laplace pressure:

$$-p_{rr}\big|_{\text{inside}} = T_1\left(\frac{1}{R_1}+\frac{1}{R_2}\right).$$

The radial normal stress inside is $-p_{rr} = p + \delta p - \tau_{rr}$, with $\tau_{rr} = 2\mu_{\text{eff}}\,\partial u_r/\partial r$. This is precisely the viscous term already present in BC3 (33), now with $\mu \to \mu_{\text{eff}}$. The remaining part $N_1 = \tau_{rr} - \tau_{\theta\theta}$ measures the *anisotropy* of the stress tensor, but it does not enter BC3 at $O(\epsilon)$. The reason is that the base state is a quiescent sphere with $\boldsymbol{\tau}_0 = 0$, so the $O(\epsilon)$ tilt of the outward normal from $\hat{\mathbf{r}}$ introduces only an $O(\epsilon)$ correction to the normal direction times an $O(\epsilon^0) = 0$ base tangential traction, contributing nothing at $O(\epsilon)$. Therefore N_1 contributes nothing extra to BC3 at $O(\epsilon)$.
At $O(\epsilon^2)$, the surface is deformed and the mismatch between normal and tangential directions at the perturbed surface generates an $O(\epsilon^2)$ correction from N_1 to the effective surface tension. This is an open problem noted in Section 12.9.

The normal stress boundary condition is:

$$(l-1)(l+2)\frac{T_1}{\rho R} = P_0 - 2\nu_{\text{eff}}(\sigma)\,\sigma\left[\,U'(1) - 2U(1)\,\right]. \tag{BC3$'$}$$

This is identical to the Newtonian BC3 (33) with $\nu \to \nu_{\text{eff}}$ from (51).

12.4 Modified governing ODE

12.4.1 The modified wavenumber q_*

Definition and derivation of q_*^2

In Reid's Newtonian derivation, the parameter $q^2 = \sigma R^2/\nu$ emerged when the r-momentum equation was multiplied through by $R/(\nu\sigma)$. For Oldroyd-B, the same step with $\nu \to \nu_{\text{eff}}$ gives:

$$q_*^2 \;\equiv\; \frac{\sigma R^2}{\nu_{\text{eff}}(\sigma)} = q^2\,\frac{1-\lambda_1\sigma}{1-\lambda_2\sigma}. \tag{52}$$

Using (50) to express $\lambda_j\sigma$ in terms of q^2 and α^2:

$$\boxed{q_*^2 = q^2 \cdot \frac{\alpha^2 - De_1\,q^2}{\alpha^2 - De_2\,q^2}.} \tag{53}$$

In both parameterisations:

$$q_*^2 = q^2 \cdot \frac{\alpha^2 - De_1\,q^2}{\alpha^2 - \beta_s De_1\,q^2}. \tag{54}$$

Structural difference from the Maxwell model

For the Maxwell fluid, $q_*^2 = q^2(1-De_1q^2/\alpha^2) = q^2(\alpha^2-De_1q^2)/\alpha^2$: the denominator is the constant α^2. For Oldroyd-B, the denominator is $\alpha^2 - De_2q^2$, which *vanishes* at $q^2 = \alpha^2/De_2$, i.e. at $\sigma = \sigma_{l;0}/De_2 = 1/\lambda_2$. As $q^2 \to \alpha^2/De_2$: $q_*^2 \to \infty$. This is the **retardation pole**: a singularity in the effective wavenumber absent from Maxwell or Newtonian problems. All qualitative differences between Oldroyd-B and Maxwell originate in this pole.

Limits of q_*^2

- $De_2 = 0$ (Maxwell): $q_*^2 = q^2(\alpha^2 - De_1q^2)/\alpha^2$. Retardation pole disappears.
- $De_1 = De_2$ (Newtonian): numerator equals denominator, $q_*^2 = q^2$.
- $q^2 = \alpha^2/De_1$ (relaxation zero): numerator $= 0$, $q_*^2 = 0$. At $\sigma = 1/\lambda_1$ the polymer is fully relaxed; it contributes zero stress, as if absent.
- $q^2 = \alpha^2/De_2$ (retardation pole): denominator $= 0$, $q_*^2 \to \infty$. At $\sigma = 1/\lambda_2$ the effective viscosity vanishes and the Bessel argument diverges.

Since $De_2 < De_1$, the retardation pole $\alpha^2/De_2 > \alpha^2/De_1$ lies to the right of the relaxation zero on the positive real q^2-axis.

12.4.2 The ODE for $U(x)$ under Oldroyd-B

The pressure field is unchanged

The pressure perturbation satisfies $\nabla^2(\delta p/\rho) = 0$ regardless of ν_{eff}. To see this: take the divergence of the linearised momentum equation $-\sigma \boldsymbol{u} = -\nabla(\delta p/\rho) + \nu_{\text{eff}} \nabla^2 \boldsymbol{u}$ and use incompressibility $\nabla \cdot \boldsymbol{u} = 0$. Since ν_{eff} is a scalar constant (it depends on σ but not on $\mathbf{x}$), we get $\nabla^2(\delta p/\rho) = \nu_{\text{eff}} \nabla^2(\nabla \cdot \boldsymbol{u}) = 0$. The solution $\delta p/\rho = \epsilon P_0 x^l Y_l^m$ is unchanged.

Following Section 5 with the substitution $\nu \to \nu_{\text{eff}}$ and hence $q^2 \to q_*^2$:

ODE for $U(x)$ under Oldroyd-B

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{l(l+1)}{x^2} + q_*^2\right] U(x) = q_*^2\, \Pi_0\, x^{l+1}, \tag{55}$$

with q_*^2 given by (53). The general solution is:

$$U(x) = C\, x\, j_l(q_* x) + \Pi_0\, x^{l+1}. \tag{56}$$

This is Reid's ODE and general solution (24)–(27) with the single replacement $q \to q_*$.

12.5 The Oldroyd-B characteristic equation

12.5.1 The ratio cancellation

Before deriving the characteristic equation, we establish a key algebraic fact that determines its structure.

The ratio α_*^2/q_*^2 is Newtonian

Define, in analogy with (34):

$$\alpha_*^2 \;\equiv\; \frac{\sigma_{l;0} R^2}{\nu_{\text{eff}}} = \alpha^2\, \frac{1 - \lambda_1 \sigma}{1 - \lambda_2 \sigma} = \alpha^2 \cdot \frac{\alpha^2 - De_1\, q^2}{\alpha^2 - De_2\, q^2}. \tag{57}$$

Computing the ratio α_*^2/q_*^2:

$$\frac{\alpha_*^2}{q_*^2} = \frac{\alpha^2 \cdot \dfrac{\alpha^2 - De_1 q^2}{\alpha^2 - De_2 q^2}}{q^2 \cdot \dfrac{\alpha^2 - De_1 q^2}{\alpha^2 - De_2 q^2}} = \frac{\alpha^2}{q^2}.$$

The Oldroyd-B factors cancel exactly in the ratio. Therefore:

$$\frac{\alpha_*^4}{q_*^4} = \frac{\alpha^4}{q^4}. \tag{58}$$

Why the cancellation occurs

The ratio $\alpha^4/q^4 = \sigma_{l;0}^2/\sigma^2$ is a pure ratio of frequencies. It is independent of viscosity by construction, since both $\alpha^2 \propto 1/\nu$ and $q^2 \propto 1/\nu$ rescale identically when $\nu \to \nu_{\text{eff}}$. The cancellation (58) holds for *any* constitutive model that linearises to $\boldsymbol{\tau} = 2\mu_{\text{eff}}(\sigma)\,\mathbf{e}$ with a scalar μ_{eff}. It is a universal feature of the linearised problem, not specific to Oldroyd-B.

12.5.2 Eliminating T_1

Modified rescaling

Multiplying BC3′ (BC3′) through by lR^2/ν_{eff}^2 and using $(l-1)(l+2)T_1/(\rho R) = \sigma_{l;0}^2 R^2/l$:

$$\underbrace{\frac{\sigma_{l;0}^2 R^4}{\nu_{\text{eff}}^2}}_{\alpha_*^4} = \underbrace{\frac{\sigma^2 R^4}{\nu_{\text{eff}}^2}}_{q_*^4}\Pi_0 - 2l\,\underbrace{\frac{\sigma R^2}{\nu_{\text{eff}}}}_{q_*^2}[U'(1) - 2U(1)]\,.$$

Hence:

$$\alpha_*^4 = q_*^2\left\{\, q_*^2\Pi_0 - 2l[U'(1) - 2U(1)]\right\}. \tag{59}$$

T_1 and ρ have cancelled completely. By (58), $\alpha_*^4 = q_*^4 \cdot \alpha^4/q^4$, so the left-hand side of the characteristic equation will be α^4/q^4, identical to the Newtonian case.

12.5.3 Applying BC1 and BC2 and assembling the characteristic equation

The algebra of Section 7.2 carries through verbatim with $q \to q_*$:

BC1: $U(1) = C\,j_l(q_*) + \Pi_0 = -1.$

BC2: Evaluating (29) on (56):

$$C\left[(2l - q_*^2)\,j_l(q_*) - 2q_*\,j_l'(q_*) + 2(l^2-1)\,j_l(q_*)\right] + 2(l^2-1)\Pi_0 = 0.$$

Solving for C and Π_0: Substituting $\Pi_0 = -1 - Cj_l(q_*)$:

$$C = \frac{2(l-1)(l+1)}{(2l - q_*^2)\,j_l(q_*) - 2q_*\,j_l'(q_*)}. \tag{60}$$

Using the Bessel derivative identity $q_* j_l'(q_*)/j_l(q_*) = l - q_* Q_{l+1/2}(q_*)$ and the Bessel recurrence to simplify, then substituting C, Π_0, $U'(1)$, and $U(1)$ into (59):

Oldroyd-B Characteristic Equation

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q_*^2}\left[l + (l+1)\,\frac{q_* - 2l\,Q_{l+1/2}(q_*)}{q_* - 2\,Q_{l+1/2}(q_*)} \right], \tag{61}$$

where

$$q_*^2 = q^2 \cdot \frac{\alpha^2 - De_1\,q^2}{\alpha^2 - De_2\,q^2} = q^2 \cdot \frac{\alpha^2 - De_1\,q^2}{\alpha^2 - \beta_s De_1\,q^2}, \tag{62}$$

$Q_{l+1/2}(q_*) = J_{l+3/2}(q_*)/J_{l+1/2}(q_*)$, and the parameters $\alpha^2 = \sigma_{l;0}R^2/\nu$, $De_1 = \lambda_1\sigma_{l;0}$, $De_2 = \lambda_2\sigma_{l;0} = \beta_s De_1$ are real constants.

Limits: $De_1 = 0$ (or $\beta_s = 1$): $q_* = q$, recovers Reid (39).
$De_2 = 0$: recovers the Maxwell characteristic equation.
$De_1 = De_2 \neq 0$: $q_* = q$, again recovers Reid (39).

This equation coincides with Zrnić & Brenn [10] Eqs. (3.11)–(3.12).

What changed from Reid's equation

1. The left-hand side $\alpha^4/q^4 + 1$ is *identical* to Reid's. (Consequence of the ratio cancellation (58).)
2. The right-hand side is Reid's right-hand side with $q \to q_*$, where q_* is the rational function (62).
3. The entire Oldroyd-B physics is encoded in De_1 and β_s (two extra real parameters beyond Reid's α^2).
4. The rational structure of q_*^2 introduces a new pole at $q^2 = \alpha^2/De_2$ with no Newtonian or Maxwell analogue.

12.6 Structure of solutions

12.6.1 The characteristic equation as a condition in q^2

Let $w = q^2$. Define the rational map:

$$R(w) \;\equiv\; q_*^2 = w \cdot \frac{\alpha^2 - De_1 w}{\alpha^2 - De_2 w}. \tag{63}$$

Then (61) is a transcendental equation $F(w) = 0$ in the single complex variable w, with α^2, De_1, De_2 as real parameters. Its roots w_j give eigenvalues $\sigma_j = w_j\nu/R^2$.

12.6.2 Two types of poles in $F(w)$

The function $F(w)$ has poles wherever the right-hand side of (61) diverges. There are two distinct sources:

Type 1: Bessel poles. The right-hand side of (61) has poles wherever $J_{l+1/2}(q_*) = 0$, i.e. at the zeros $z_1^* < z_2^* < \cdots$ of the Bessel function in the q_*-plane. These satisfy $q_* = z_n^*$, i.e. $R(w) = (z_n^*)^2$: for each n, this is a quadratic (in w) algebraic equation with up to two solutions in the w-plane. For small De_1 the rational map $R(w)$ is a small perturbation of the Newtonian case, and a heuristic application of the argument principle suggests that $F(w) = 0$ retains exactly two roots per strip, as in the Newtonian problem; a rigorous contour argument for general De_1 is open (see §12.9). The first strip ($n = 1$) captures the **dominant pair**, governing the fundamental capillary oscillation.

Type 2: Retardation pole. The factor $\alpha^2 - De_2 w$ in the denominator of $R(w)$ vanishes at:

$$w_{\text{ret}} = \frac{\alpha^2}{De_2}, \qquad \sigma_{\text{ret}} = \frac{\sigma_{l;0}}{De_2} = \frac{1}{\lambda_2}. \tag{64}$$

As $w \to w_{\text{ret}}$, $q_* \to \infty$ and $Q_{l+1/2}(q_*)$ oscillates rapidly. Near this singularity the characteristic equation (61) can be satisfied, capturing one or two **retardation roots**. These correspond to modes decaying at rate $\sim 1/\lambda_2$. Since $\lambda_2 < \lambda_1$, the retardation mode always decays faster than the Maxwell-type relaxation mode; whether it is faster or slower than the fundamental capillary pair depends on De_2.

Physical meaning of the retardation roots

At $\sigma = 1/\lambda_2$, the solvent and polymer contributions to the tangential stress cancel: $\mu_{\text{eff}}(1/\lambda_2) = 0$. The retardation roots describe a mode in which the polymer network, instead of resisting the flow, briefly *drives* it, before the solvent dissipates the energy. This phenomenon has no Newtonian analogue; it requires two fluid components. Physically, it corresponds to the elastic recoil of the polymer network overshooting the equilibrium and being damped by the solvent on the retardation timescale λ_2.

12.6.3 Dominance condition for the fundamental pair

The fundamental pair of roots governs observable drop oscillation provided the retardation roots are faster-decaying (subdominant):

$$\text{Re}(\sigma_{1,2}) < \frac{1}{\lambda_2} \quad \Longleftrightarrow \quad De_2 \cdot \text{Re}(\beta_{1,2}) < 1, \tag{65}$$

where $\beta_j = q_j^2/\alpha^2 = \sigma_j/\sigma_{l;0}$.

When (65) holds, the retardation roots are higher overtones: they decay faster than the capillary oscillation period and are irrelevant to the Moláček & Bush quasi-static model. When (65) fails, the retardation roots must be retained.

12.7 A_l and D_l for the Oldroyd-B fluid

12.7.1 Vieta extraction: the two-number picture survives

Provided (65) holds, the dynamics of mode l is governed by the two roots $\beta_{1,2}$ of the characteristic equation (61) in the first Bessel strip. The Vieta extraction of Section 9 applies without modification:

$$A_l^{\mathrm{OB}} = \frac{1}{\beta_1\beta_2}, \qquad D_l^{\mathrm{OB}} = \frac{\beta_1+\beta_2}{2a\,\beta_1\beta_2}, \qquad a = \mathrm{Oh}\sqrt{l(l-1)(l+2)}. \tag{66}$$

The difference from the Newtonian case is that $\beta_{1,2}$ are now roots of (61) rather than (39): they depend on (De_1, β_s) in addition to (l, Oh).

Parameter dependence of A_l^{OB} and D_l^{OB}

$$A_l^{\mathrm{OB}} = A_l^{\mathrm{OB}}(l,\,\mathrm{Oh},\,De_1,\,\beta_s), \qquad D_l^{\mathrm{OB}} = D_l^{\mathrm{OB}}(l,\,\mathrm{Oh},\,De_1,\,\beta_s). \tag{67}$$

Limits:

- $De_1 \to 0$ (or $\beta_s \to 1$): recover Moláček & Bush Newtonian values.
- $\beta_s \to 0$ at fixed De_1: recover Maxwell values (a special case of the Oldroyd-B framework).
- $De_1 \to \infty$ at fixed β_s: strongly elastic limit; both coefficients must be determined numerically.

Monotone inequalities between models (conjectured)

For the same total viscosity $\mu = \mu_s + \mu_p$ and the same Oh, numerical evidence suggests:

$$A_l^{\mathrm{N}} \le A_l^{\mathrm{OB}} \le A_l^{\mathrm{M}}, \qquad D_l^{\mathrm{M}} \le D_l^{\mathrm{OB}} \le D_l^{\mathrm{N}}.$$

Elasticity enhances effective inertia (increases A_l) and reduces instantaneous dissipation (decreases D_l). The solvent, present in Oldroyd-B but absent in Maxwell, partially restores both toward their Newtonian values. The interpolation appears monotone in β_s: as β_s increases from 0 (Maxwell) to 1 (Newtonian), A_l decreases and D_l increases. These inequalities are consistent with physical intuition but have not been established analytically from the transcendental characteristic equation; a formal proof is open.

12.7.2 When the two-number picture breaks

When condition (65) fails, the retardation roots have decay rates comparable to the fundamental pair. There are then three (or more) roots of comparable magnitude, and the Moláček & Bush quadratic ODE for b_l is no longer adequate. In capillary time $\tau = \sigma_{l;0}\, t$

(all quantities dimensionless), the full description is an integro-differential equation,

$$A_l^{\mathrm{OB}}\,\ddot{b}_l + 2a\int_0^{\tau} K^{(\mathrm{cap})}(\tau-\tau')\,\dot{b}_l(\tau')\,\mathrm{d}\tau' + b_l = 0, \qquad a \equiv \mathrm{Oh}\sqrt{l(l-1)(l+2)}, \tag{68}$$

where dots denote $\mathrm{d}/\mathrm{d}\tau$ and the dimensionless OB memory kernel in capillary time is

$$K^{(\mathrm{cap})}(\tau) = \beta_s\,\delta(\tau) + \frac{1-\beta_s}{De_1}\,e^{-\tau/De_1}. \tag{69}$$

Its positive-exponent Laplace transform is $\tilde{K}^{(\mathrm{cap})}(\beta) = \int_0^\infty K^{(\mathrm{cap})}(\tau)\,e^{\beta\tau}\,\mathrm{d}\tau = \nu_{\mathrm{eff}}(\beta\,\sigma_{l;0})/\nu = (1-De_2\,\beta)/(1-De_1\,\beta)$, and its zeroth moment is $\int_0^\infty K^{(\mathrm{cap})}\,\mathrm{d}\tau = \beta_s + (1-\beta_s) = 1$. In the Newtonian limit ($De_1 \to 0$, or equivalently $\beta_s \to 1$): $K^{(\mathrm{cap})} \to \delta(\tau)$, and the equation reduces to $A_l^{\mathrm{OB}}\ddot{b} + 2a\dot{b} + b = 0$. This is a structured approximation. The convolution kernel replaces the full Newtonian damping function $D_l(\mathrm{Oh})$ by the zeroth moment $\tilde{K}^{(\mathrm{cap})}(0) = 1$. It captures the frequency-dependent viscosity $\nu_{\mathrm{eff}}(\beta\sigma_{l;0})/\nu$ but not the Oh-dependence in Reid's Bessel-function structure. The exact $D_l(\mathrm{Oh})$ is obtained via Vieta extraction from (61) at $De_1 = 0$. The dimensional kernel $K_{\mathrm{OB}}(s)$ satisfies $\tilde{K}_{\mathrm{OB}}(\sigma) = \int_0^\infty K_{\mathrm{OB}}(s)\,e^{\sigma s}\,\mathrm{d}s = \nu_{\mathrm{eff}}(\sigma)/\nu$:

$$K_{\mathrm{OB}}(s) = \beta_s\,\delta(s) + \frac{1-\beta_s}{\lambda_1}\,e^{-s/\lambda_1}, \tag{70}$$

related to the capillary kernel by $K^{(\mathrm{cap})}(\tau) = \sigma_{l;0}^{-1}\,K_{\mathrm{OB}}(\tau/\sigma_{l;0})$. The δ term encodes the instantaneous **solvent** response with weight $\beta_s = \mu_s/\mu$; the exponential term encodes the polymer memory with weight $(1-\beta_s) = \mu_p/\mu$ and relaxation time λ_1. The two-number approximation corresponds to replacing $K^{(\mathrm{cap})}$ by its zeroth moment $= 1$ (treating memory as instantaneous), valid when $De_2 \cdot \mathrm{Re}(\beta_{1,2}) \ll 1$.

12.8 Four clean limits

Limit	Condition	q_*^2	Char. eq.	A_l, D_l
Newtonian	$De_1 = 0$	q^2	Reid (39)	Molaček & Bush
Maxwell	$\beta_s = 0$	$q^2\left(1 - \frac{De_1\,q^2}{\alpha^2}\right)$	Khismatullin & Nadim	(66), $\beta_s = 0$
Weak elasticity	$De_1 \ll 1$	$q^2[1-(De_1 - De_2)q^2/\alpha^2]$	Perturbation of (39)	Analytic correction
No retardation	$De_2 \ll De_1$	$\approx q^2(1 - De_1 q^2/\alpha^2)$	$\approx$ Maxwell	$\approx$ Maxwell values

12.9 Open problems and references

What is established.

1. The characteristic equation (61)–(62) is exact at $O(\epsilon)$ for a free-surface Oldroyd-B drop in air.

2. The ratio cancellation (58) holds for any linearisable constitutive model, making the left-hand side of the characteristic equation universal.

3. The dominance condition (65) gives an explicit criterion for when the two-number $\{A_l, D_l\}$ extraction is valid.

4. The integro-differential ODE (68)–(70) is the natural viscoelastic extension of the Molaček & Bush quadratic, capturing the frequency-dependent damping ν_{eff}/ν when the two-number picture breaks down.

What is open.

1. **Numerical tabulation.** A complete table of $A_l^{\text{OB}}(l, \text{Oh}, De_1, \beta_s)$ and $D_l^{\text{OB}}(l, \text{Oh}, De_1, \beta_s)$ analogous to Molaček & Bush Table 1 does not exist in the literature. The Newton's method strategy of Section 8 applies directly: for each (De_1, β_s), initialise from the Newtonian roots and track them as (De_1, β_s) are increased.

2. **Large De_1 root structure.** For $De_1 \gg 1$, the numerator $\alpha^2 - De_1 q^2$ changes sign for real $q^2 > \alpha^2/De_1$, making q_*^2 negative there. The physical consequence is that Bessel functions $j_l(q_* x)$ become modified spherical Bessel functions $i_l(|q_*|x)$, changing the radial profile from oscillatory to monotone. A complete argument-principle analysis for large De_1 does not exist.

3. **Second-order normal stress correction.** The $O(\epsilon^2)$ contribution from $N_1 = \tau_{rr} - \tau_{\theta\theta}$ modifies the effective surface tension as $T_1 \to T_1^{\text{eff}} = T_1\left[1 + \tilde{c}\,\epsilon\,\text{Re}(N_1^{(2)} R/T_1)\right]$ with $\tilde{c}$ a dimensionless $O(1)$ coefficient (the precise value requires the second-order calculation). For $\epsilon = O(0.1\text{–}0.3)$ (typical bouncing-drop experiments), this correction could reach several percent in $\sigma_{l;0}$.

4. **Experimental validation.** No experiment has directly measured A_l and D_l for a viscoelastic drop and compared to the characteristic equation (61). PEO-water drops in the range $De_1 \in [0.1, 2]$ and $\beta_s \in [0.5, 0.9]$ would be the natural first target.

Key references.

- **Oldroyd (1950)** [6]: original derivation of (44); clearest source for the physical meaning of λ_1, λ_2, and β_s.

- **Bird, Armstrong & Hassager (1987)** [9]: dumbbell-model derivation of Oldroyd-B; physical meaning of $\beta_s = \mu_s/(\mu_s + \mu_p)$.

- **Khismatullin & Nadim (2001)** [7]: shape oscillations of a viscoelastic drop including the full Oldroyd/Jeffreys (relaxation + retardation) constitutive law; their characteristic equation already covers the Oldroyd-B case.

- **Zrnić & Brenn (2024)** [10]: free-surface Oldroyd-B drop oscillations; their Eqs. (3.11)–(3.12) are the same characteristic equation and q_*^2 derived in §12. The present notes expand the derivation and connect it to the Molaček & Bush two-number framework.

- **Mukherjee & Sarkar (2010)** [8]: Oldroyd-B drop in a Newtonian bath; numerical results for oscillation frequency and damping as functions of De_1 and β_s, usable as benchmark for the free-surface limit.

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